# Zhican Zhou

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## Personal Particulars

I am currently a Research Fellow at the National University of Singapore (NUS). I received my Ph.D. in Electrical and Computer Engineering (ECE) from King Abdullah University of Science and Technology (KAUST). My research focuses on silicon photonics, hybrid optoelectronic computing systems, hardware-aware neural-network design, and privacy-preserving hardware architectures. I have authored or co-authored **23 journal and conference papers**, including key works published in ***Nature Electronics*** and ***Nature Communications***. These works have attracted broad media attention, with coverage by EurekAlert!, Tech Xplore and Optics & Photonics News, as well as Saudi national television. I have gained experience in graduate-level teaching and research mentoring by serving as a teaching assistant for two ECE courses and mentoring six research interns. I have also contributed to multiple funded research projects, including proposal preparation, experimental planning, and project execution. In recognition of my contributions to integrated photonic computing and hardware-rooted security, I was awarded the **2026 Tingye Li Memorial Scholarship**, a prestigious international award presented to only two recipients worldwide each year.

## Academic Experience & Awards

**National University of Singapore** — Research Fellow — 2026.10 – Present
- Department of Electrical and Computer Engineering — Singapore

**King Abdullah University of Science & Technology** — Ph. D. — 2022.9 – 2026.9
- Department of Electrical and Computer Engineering — Jeddah, Saudi Arabia

> **University of Texas at Austin** — Visiting Student — 2024.4 – 2024.7
> - Chandra Department of Electrical and Computer Engineering — Austin, USA
>
> **Peking University** — Visiting Student — 2024.9 – 2024.12
> - Photonic Chip and Information System Innovation Center — Beijing, China

**Nankai University & National Center for Nanoscience and Technology, China (NCNST)** — Joint M.Sc. — 2019.9 – 2022.6
- Photonics & Photonic Technology, top 5% of class — Beijing, China

**Chongqing University** — B. Eng — 2015.9 – 2019.6
- Materials science and technology, Top 10% of class — Chongqing, China

**Awards:** 2026 Tingye Li Memorial Scholarship (2 recipients worldwide each year), Excellence Master Graduation Thesis in Tianjin (Top 1%), NCNST Outstanding Graduate (Top 1%), KAUST CEMSE Dean's Award (Top 20%), Nankai University Outstanding M.Sc. (Top 5%), Chongqing University Outstanding Undergraduate (Top 10%).

## Project Experience

**Principal contributed student:**

Served as a core graduate researcher under faculty supervision, with substantial responsibility for drafting technical sections of project proposals, developing experimental plans, executing and coordinating research activities, and preparing progress

reports, final deliverables and technical presentations.

[1] **Optoelectronic Hardware Acceleration for Variational Autoencoder,** Program from the Center of Excellence in Data Science and Artificial Intelligence (SDAIA), 01/01/2023-01/01/2024, budget US$ 53,320.

[2] **Broadband Photonics with Quantum Dot Driven Lithium Niobate Integrated Circuits,** Competitive Research Grant (CRG) in KAUST, 01/04/2025-01/04/2028, budget US$ 765,000.

[3] **Unlocking AI Acceleration: Integrated Silicon Photonics with Quantum Dot Lasers,** Basic Science Grant (BSG) from Research Development and Innovation Authority (RDIA) of Saudi Arabia, from 01/06/2025, budget US$ 350,000. # Exhibit: 2025 Future Economic Innovation Week, Riyadh (student representative)

**Participating student:**

Contributed to selected experimental activities, including device testing and characterization, data collection and analysis, and preparation of supporting technical documentation.

[4] **Monolithic Integration for Future Mobility,** Semiconductor Initiative 2022 project, 12/01/2022 – 11/30/2023, budget US$ 162,500.

[5] **Autonomous Mining: Leveraging Silicon Photonics-Integrated LiDAR for Driverless Systems**, KAUST-Translational Research Grants (TRG), 06/30/2024 – 06/30/2027, budget US$ 1,500,000.

## Teaching Experience

**Teaching Assistant, KAUST | 2024–2025**

Assisted with course delivery by answering students' technical questions, holding office hours and review sessions, preparing instructional materials, grading assignments and examinations, and supporting examination administration.

- ECE 300: Advanced Topics in Integrated Photonics —Advanced doctoral-level course, ~10 students Spring Semester, 2024
- ECE 208: Semiconductor Optoelectronic Devices — ECE M.Sc. core course, ~20 students Fall Semester, 2025

**Student Mentoring, KAUST | 2023–2026**

Served as a graduate research mentor to six undergraduate and graduate interns, providing day-to-day guidance on project development, experimental work, data analysis and technical communication.

- Nakul Nandhakumar — Undergraduate Research Intern, University of California San Diego Jun. 2024–Dec. 2024
  Mentored a six-month research project that resulted in a first-author paper presented orally at CLEO 2025. He subsequently joined the Ph.D. program at UC San Diego.
- Aleksandr Movsisian — Research Intern, Moscow State University Jul. 2025–Jun. 2026
  Provided research guidance throughout a one-year project, with a manuscript currently in preparation. He subsequently joined ETH Zurich as a Ph.D. student.
- Wenkai Shan —Research Intern, National University of Singapore, Oct. 2025–Dec. 2025
  Mentored a two-month project in photonic computing. He subsequently continued as a Ph.D. student at NUS.
- Yousef Khojah — KAUST Gifted Student Program, University of Illinois Urbana-Champaign, , Jul. 2025–Aug. 2025
  Mentored a short-term research project in integrated photonics. He is expected to begin his Ph.D. studies at KAUST.
- Daniele Veraldi and Ernesto G. Saleh — KAUST Photonics Summer Camp Jul. 2023–Aug. 2023
  Mentored one-month projects in optical computing. They subsequently continued graduate study at Sapienza University of Rome and the Technical University of Denmark, respectively.

## Publication & Services

### ❖ Journal Publications

[1] **Zhican Zhou**, et. al, "Physical unclonable functions based on chaotic vertical-cavity surface-emitting lasers for dynamic authentication", ***Nature Electronics***, 9, 675–685 (2026). 2025 IF = 42.3, JCR/Q1.

[2] **Zhican Zhou**, et. al, "Prospects and applications of on-chip lasers", ***eLight***, 3, 1 (2023). Cover paper, ESI highly cited paper, 2025 IF = 31.4, JCR/Q1.

[3] **Zhican Zhou**, et. al, "Integrated quantum dot lasers for parallelized photonic edge computing" ***Advanced Photonics***, 8, 026017 (2026). 2025 IF = 19.5, JCR/Q1.

[4] Xiaowen Zhang†, Xiangpeng Ou†, **Zhican Zhou**†(co-first author), et. al, "Integrated van der Waals Waveguides for All-Optical Nonlinear Photonic Circuits", ***Nature Communications***, 17, 6549 (2026). 2025 IF = 18.1, JCR/Q1.

[5] **Zhican Zhou**, et. al, "Emerging of Two-dimensional Materials in Novel Memristor", ***Frontiers of Physics***, 17, 23204 (2022). 2025 IF = 6.6, JCR/Q1.

[6] Shu Wang†, **Zhican Zhou**†(co-first author), et. al, "All-atomristor logic gates", ***Nano Research***, 16, 1688–1694 (2023). 2025 IF = 9.4, JCR/Q1.

[7] Shu Wang†, **Zhican Zhou**†(co-first author), et. al, "Progresses on new generation laser direct writing technique," ***Materials Today Nano***, 16, 100142 (2021). 2024 IF = 7.6, JCR/Q1.

[8] Maolin Chen, …, **Zhican Zhou**, et. al "Ultrawide-bandwidth boron nitride photonic memristors", ***Nature Nanotechnology***, 20, 1633–1640 (2025). 2025 IF = 37.5, JCR/Q1.

[9] Meng Wang, Xiaofeng Wang, **Zhican Zhou**, et. al, "High-spatial-resolution composition analysis of micro/nano-structures with a nanoscale compositional variation," ***Nano Research***, 16, 1090–1095 (2023). 2025 IF = 9.4, JCR/Q1.

[10] Xiangpeng Ou, …, **Zhican Zhou**, et. al, "Mode Control and Dynamic Population Gratings in Quantum-Dot Lasers." ***Optica***, 13, 469-475 (2026). 2025 IF = 8.8, JCR/Q1.

[11] Yuxuan Xie, …, **Zhican Zhou**, et. al, "Tunable Lasers Hybrid Integrated with Silicon Photonics at 2.0-μm", ***Photonic Research***, 13, 737-742 (2025). 2025 IF = 7.1, JCR/Q1.

[12] Yueqi Li, …, **Zhican Zhou**, et. al, "Nanopore/Nanosphere-Induced Optical Enhancement of Monolayer $MoS_2$." ***Advanced Optical Materials***, 11, 2300120 (2023). 2025 IF = 7.2, JCR/Q1.

[13] Fengyou Yang, …, **Zhican Zhou**, et. al, "High-performance optoelectronic memory based on bilayer $MoS_2$ grown by Au catalyst," ***Journal of Materials Chemistry C***, 8, 2664-2668 (2020). 2025 IF = 5.1, JCR/Q2.

[14] Xiaofeng Wang, …, **Zhican Zhou**, et. al, "Shape-dependent close-edge 2D-$MoS_2$ nanobelts," ***RSC Advances***, 10, 33544-33548 (2020). 2025 IF = 6.1, JCR/Q2.

[15] Junqi Wang, …, **Zhican Zhou**, et. al, "High-performance self-powered MoS1. 2Se0. 8 lateral homojunction photodetector with broadband spectrum response." ***Applied Physics Letters***, 127, 143501 (2025). 2025 IF = 3.8, JCR/Q2.

[16] Shengyao Chen, …, **Zhican Zhou**, et. al, "Laser modulation of the FePS3-based synaptic flexible memristors," ***Nanotechnology***, 36, 055201 (2024). 2025 IF = 2.9, JCR/Q3.

### ❖ Conference paper

[1] **Zhican Zhou**, et. al, "On-Chip Quantum Dot Lasers Driven High-Speed Optical Neural Networks", ***Conference on Lasers and Electro-Optics (CLEO)*** 2024, Charlotte, North Carolina, USA. (Oral Presentation)

[2] **Zhican Zhou**, et. al, "On-chip Quantum Dot Lasers for Photonic Processing Unit on Silicon-Based Heterogeneous Platform", *IEEE Photonics Conference (IPC)* 2024, Rome, Roma provincial, Italy. (Oral Presentation)

[3] Hanqing Zhu, **Zhican Zhou**, et. al, "ENLighten: Lighten the Transformer, Enable Efficient Optical Acceleration", *Asia and South Pacific Design Automation Conference (ASP-DAC)* 2026. Preprint version: arxiv.org/abs/2510.01673, Hong Kong. (Oral Presentation)

[4] Nakul Nandhakumar, **Zhican Zhou**, et. al, "Deep Learning-Enhanced Dynamic Security System Based on Photonic Physical Unclonable Functions", *Conference on Lasers and Electro-Optics (CLEO)* 2025, Long Beach, California, USA. (Oral Presentation)

[5] Lu, Hang, **Zhican Zhou**, et. al, "Optical Authentication Leveraging Chaotic-Cavity VCSEL Far-Field Patterns with Deep Learning." *IEEE Photonics Conference (IPC)* 2025, pp. 1-2. (Oral Presentation)

[6] Xiangpeng Ou, **Zhican Zhou**, et. al, "Advancing post-Moore performance scaling in Si photonics: heterogeneous integration of quantum dot lasers." *In Novel In-Plane Semiconductor Lasers XXIII* 2023, p. PC129050A. SPIE, Tokyo Bay, Japan. (Oral Presentation)

[7] Xiangpeng Ou, **Zhican Zhou**, et. al, "On-Chip Lasers in Silicon Photonics: Pathways to Integration and Applications." *IEEE Photonics Conference (IPC)* 2023, pp. 1-2. IEEE, Orlando, Florida, USA. (Oral Presentation)

❖ **Acadamic Services:**

**Peer Review:** Reviewed >10 journals/conferences as an independent reviewer/co-reviewer, including IEEE TCSVT, npj Flex. Electron., Nat. Comm., Light: Sci. & Appl., PhotoniX, Photon. Res., J. Lightwave Technol., NeurIPS.

**Academic Event Service:** Lead student organizer for the 2023 Semiconductor Photonics Workshop (100+ attendees), coordinating logistics and on-site operations; student volunteer for the 2023 KAUST Semiconductor Forum, assisting with event logistics, registration, and session support.

**Professional Society Activities (Optica):** Member, Optica Photonics Student Chapter (Saudi Arabia); participated in chapter operations and helped organize academic events (seminars, invited talks).